

CS736 - Assignment-3

Team Members

1. Atharv Shingewar (24M0834)
2. Russel Abreo (24M2107)

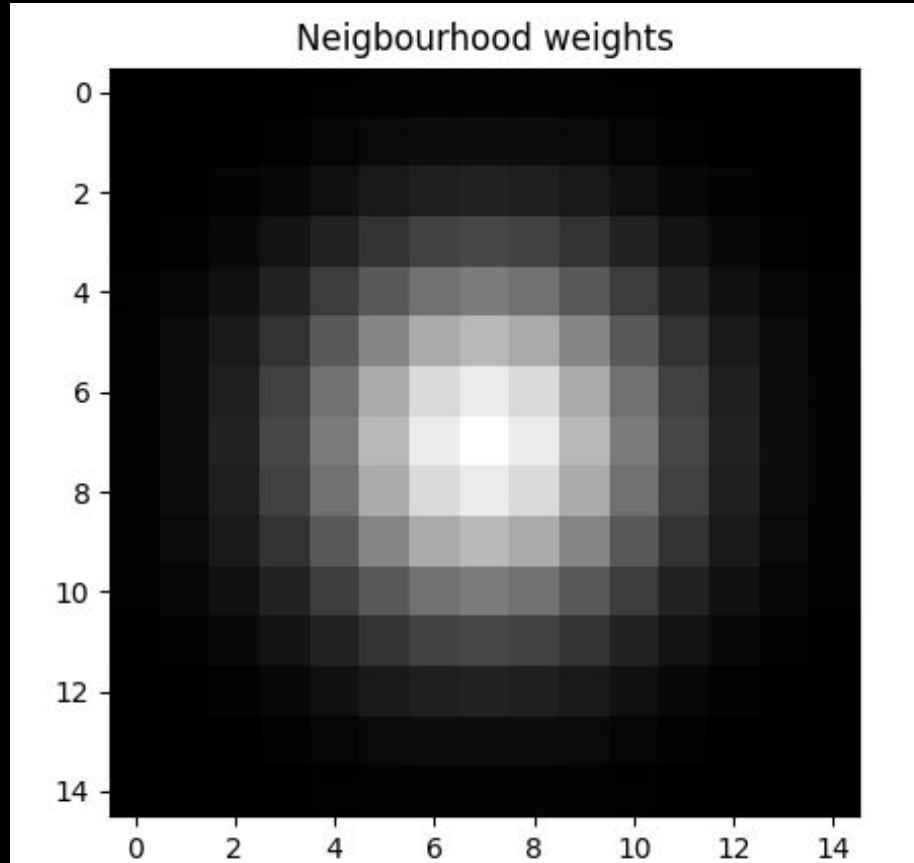
Under the Guidance of :

Prof. Suyash Awate

Q1

1. $q = 1.6$

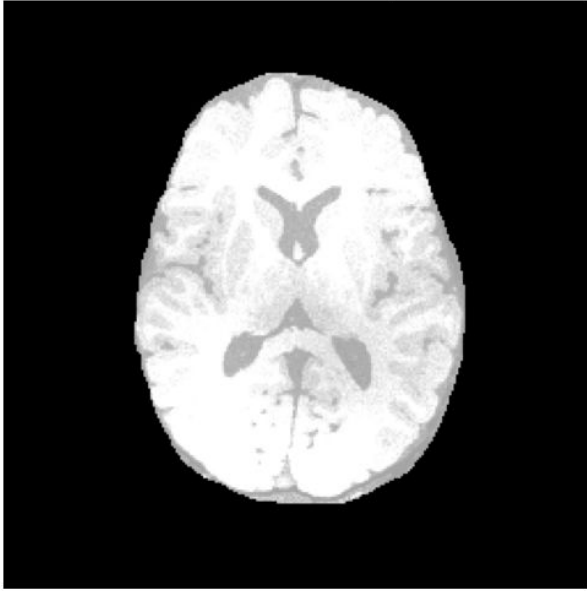
Q1 b - Neighbourhood weights



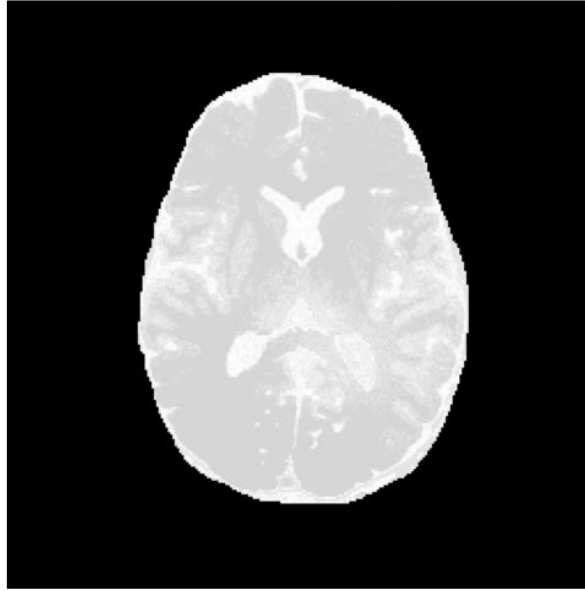
Q1 c - Initial estimates for Memberships

Initial Membership Distributions

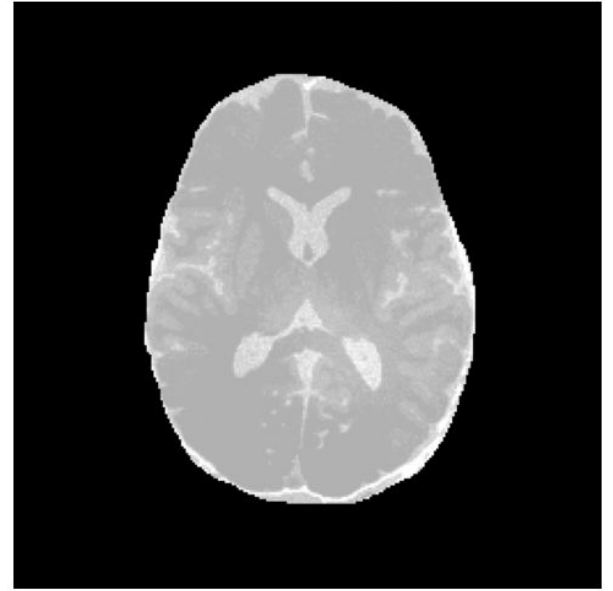
Class 1 Membership



Class 2 Membership



Class 3 Membership



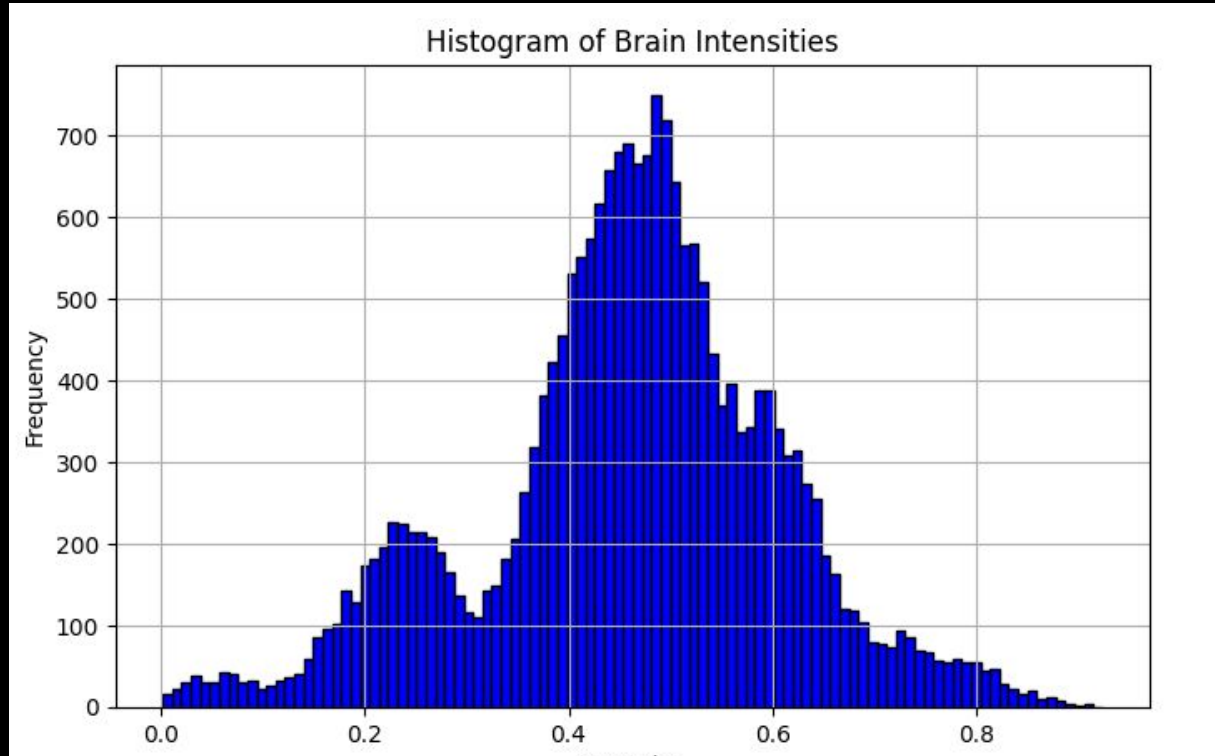
Q1 c - Initial estimates for Memberships

1. For each pixel intensity, compare its value with the three histogram peak intensities (chosen as initial class means) and assign the membership probability based on intensity similarity.
2. These similarity scores are then normalized across all class means, ensuring that the membership estimates sum to 1.

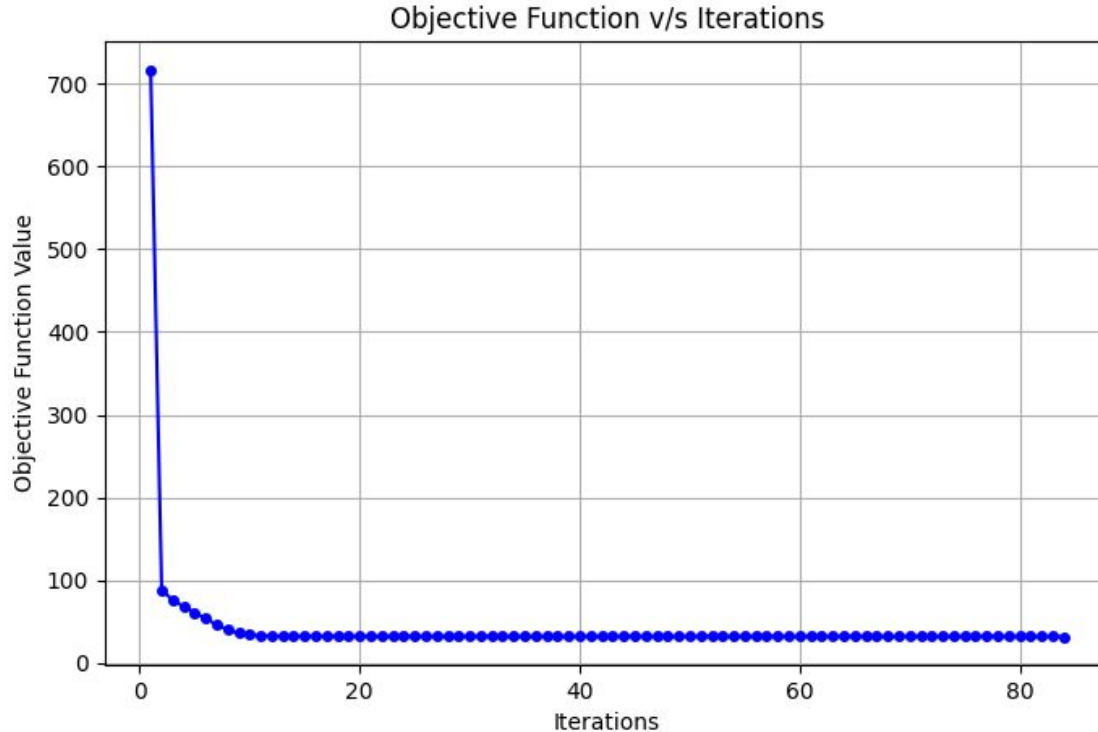
Q1 d

1. Initial class means: [set as per observations from the histogram image intensities]
0.5,
0.25,
0.1
2. Optimal class means:
0.59480953,
0.48876253,
0.26209626

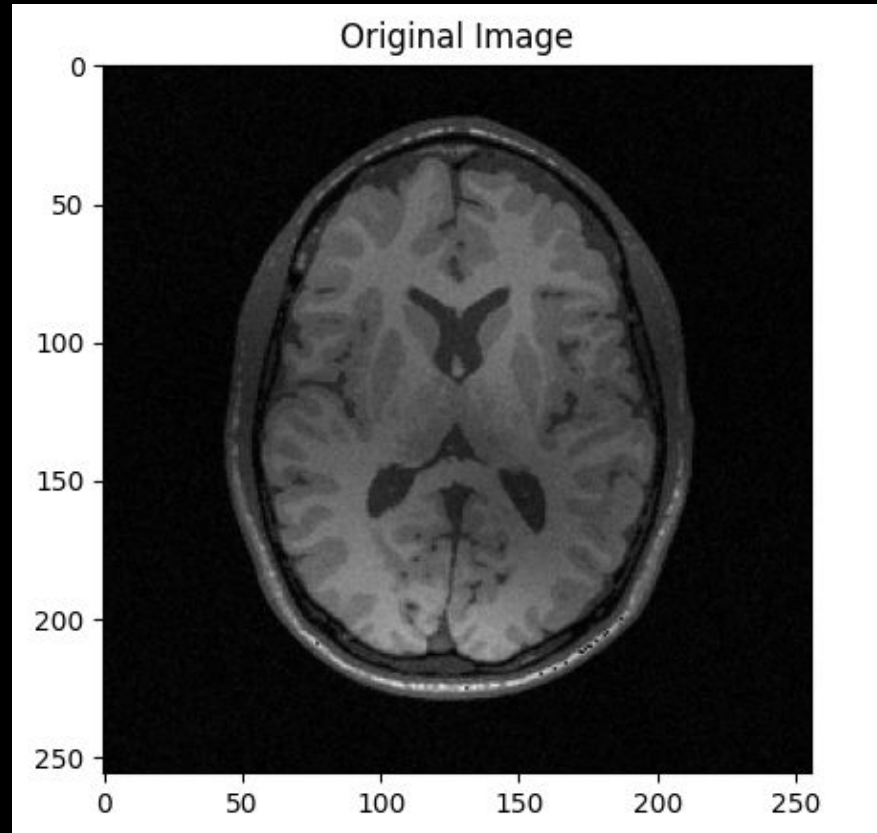
Q1 d - Histogram of Image Intensities



Q1 e - Objective Function Plot



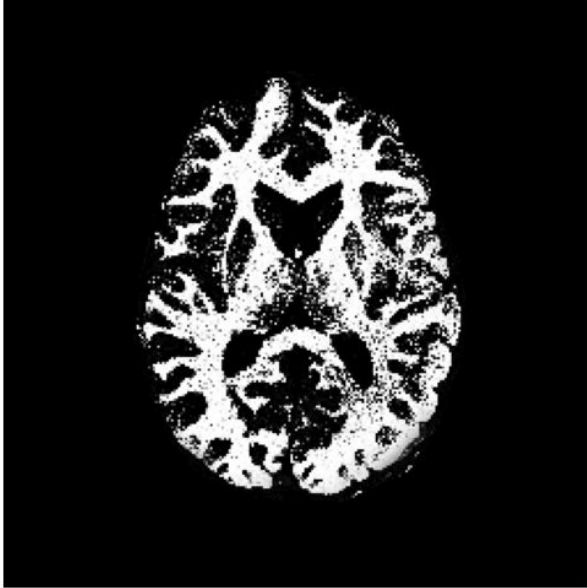
Q1 (i) - Corrupted Image



Q1 (ii)- Optimal estimates for Memberships

Optimal Membership Distributions

Class 1 Membership



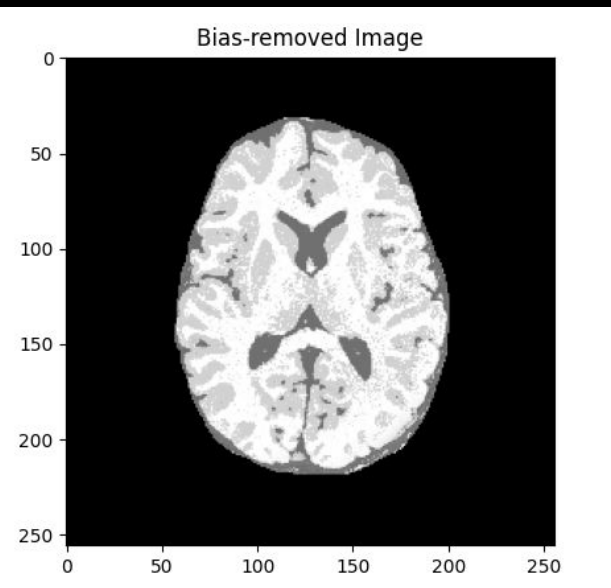
Class 2 Membership



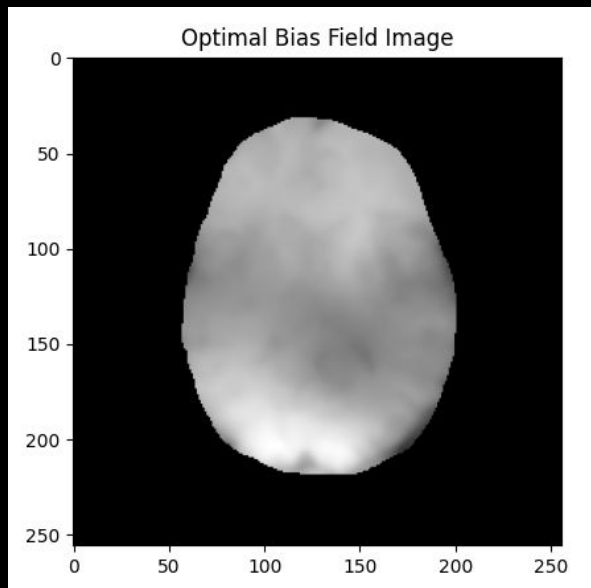
Class 3 Membership



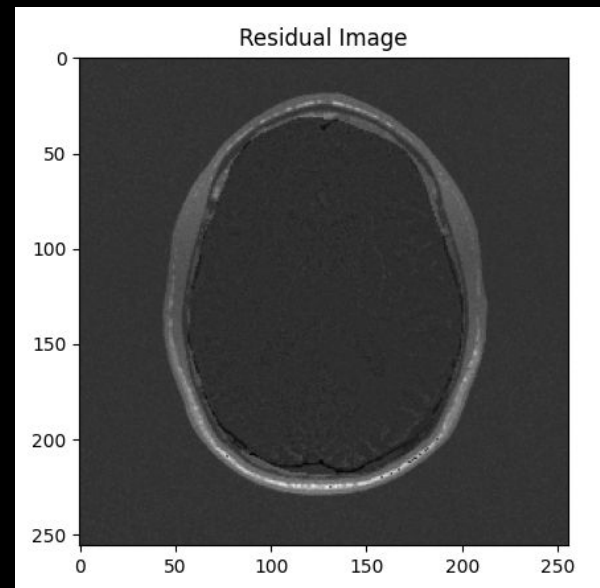
Q1(iii) - Bias Removed, Bias Field, Residual Image



Bias Removed
Image



Bias Field

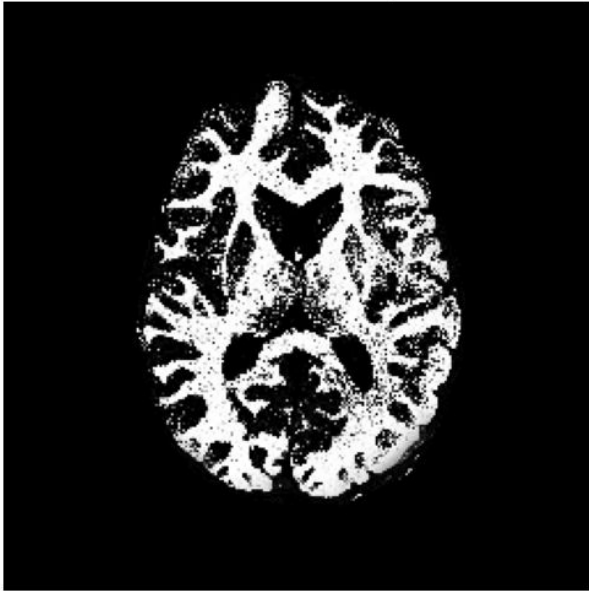


Residual Image

Q1 - Unique Solution

Optimal Membership Distributions

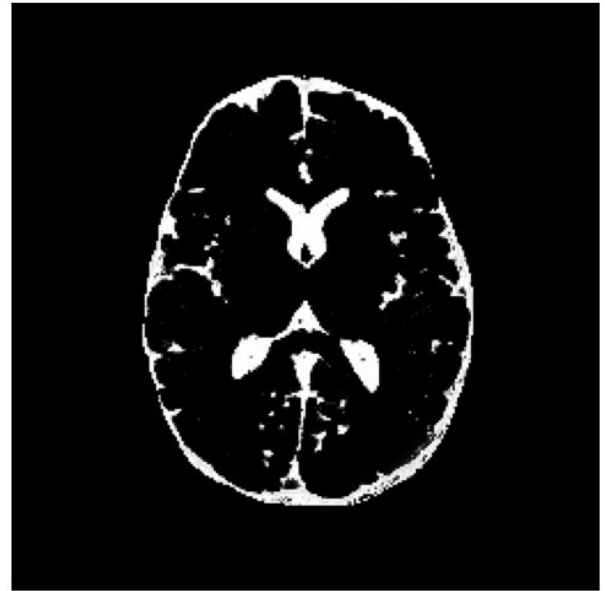
Class 1 Membership



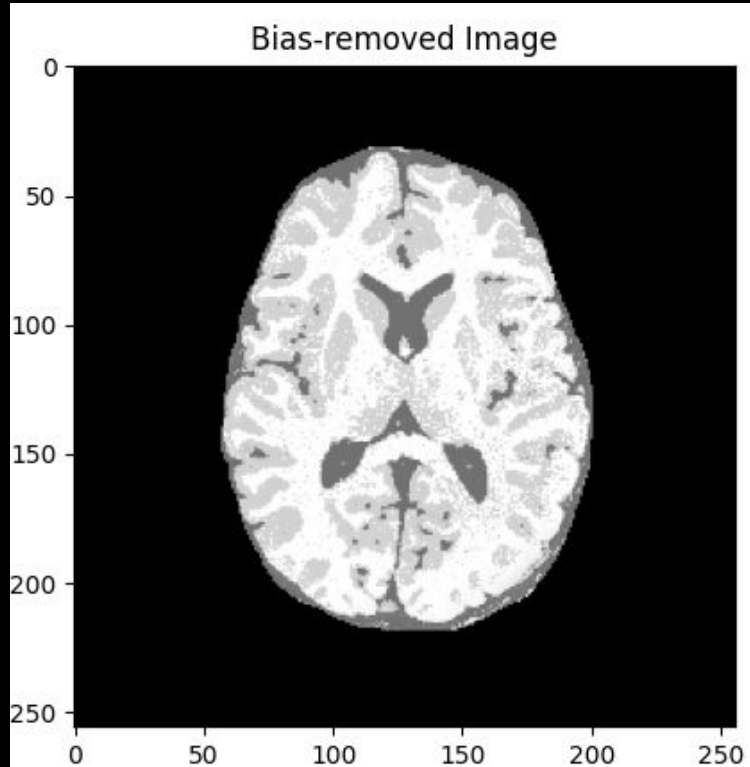
Class 2 Membership



Class 3 Membership



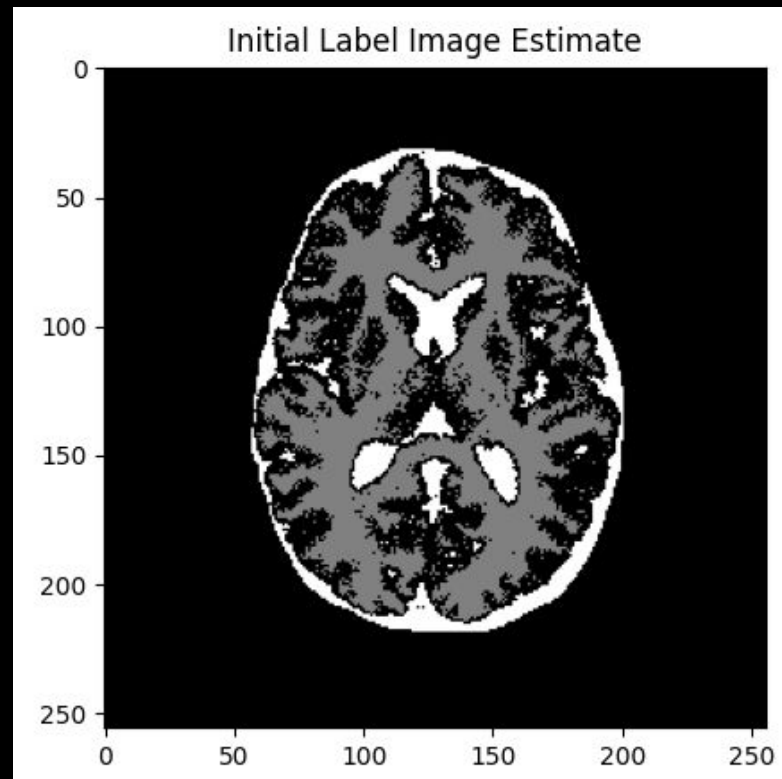
Q1 - Unique Solution



Q2 a

Chosen beta = 2

Q2 b

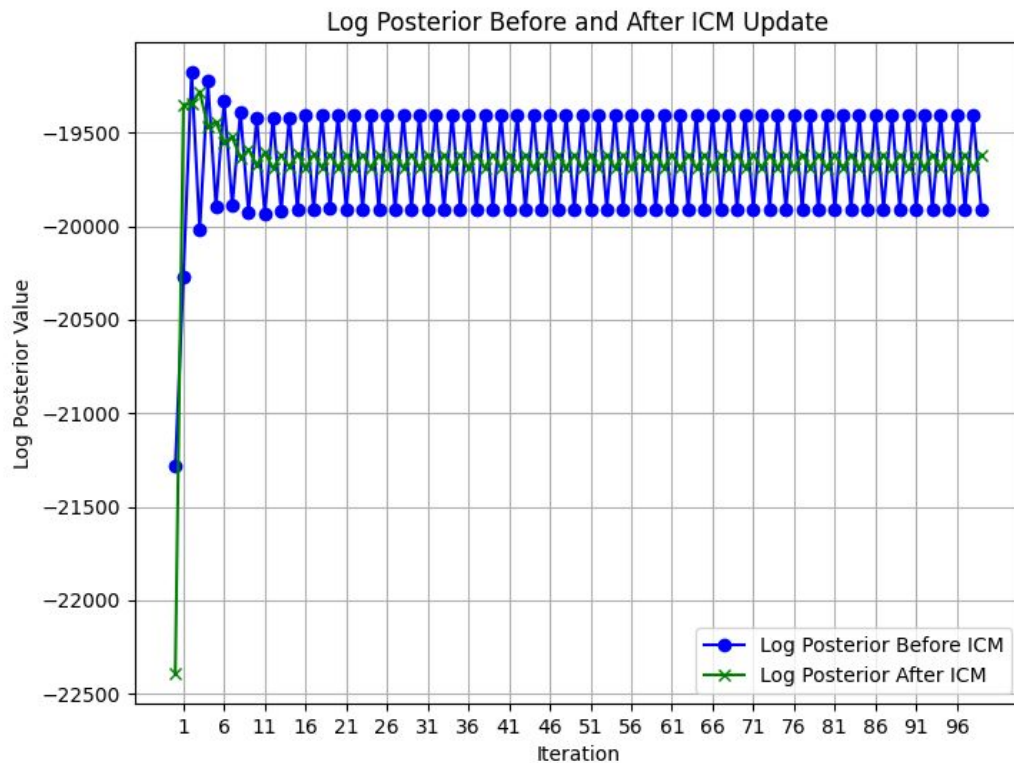


Q2 c

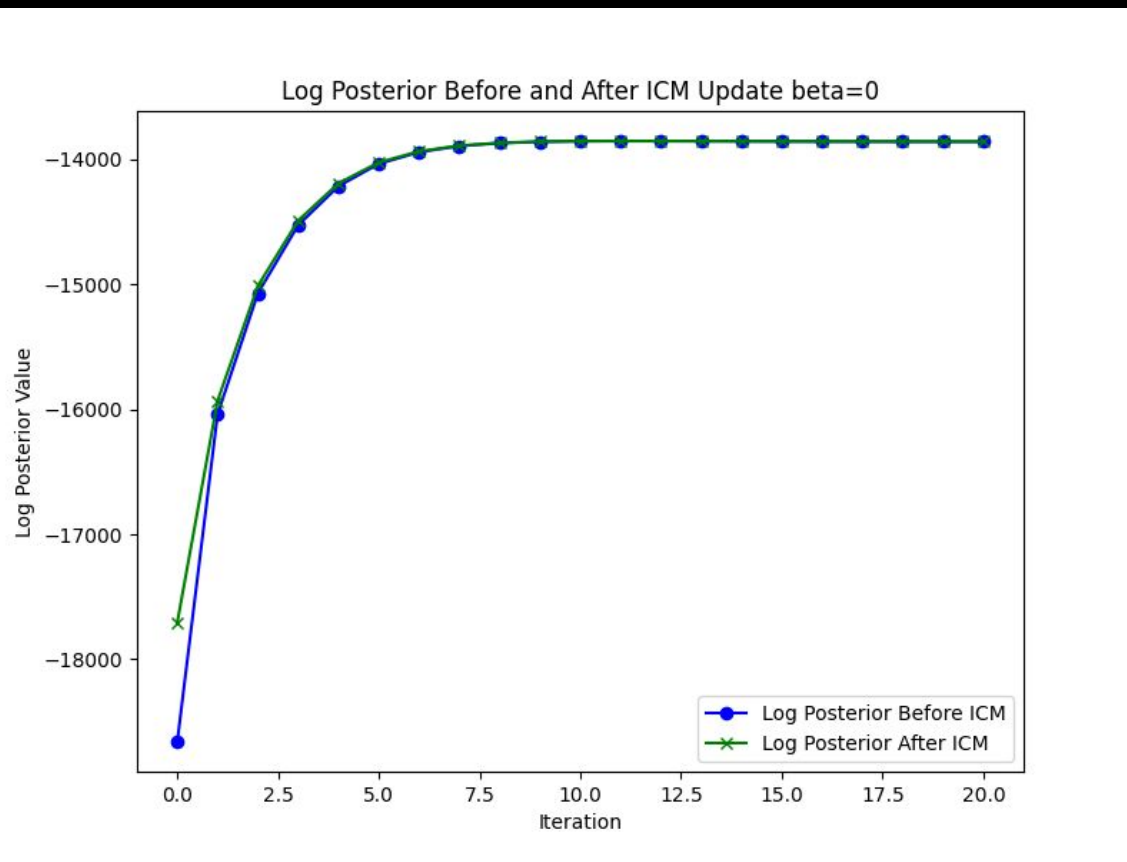
Mean = [0.50408191, 0.62782413, 0.26943405]

std = [0.04414, 0.03704, 0.0773]

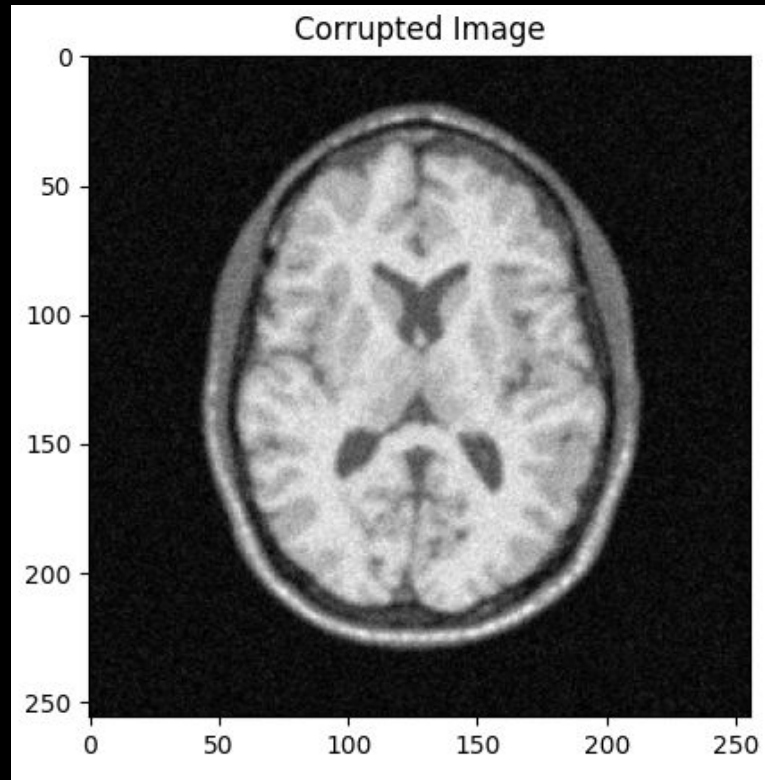
Q2 d



Q2 d



Q2e (i)



Q2e (ii)

Optimal Class Membership Estimate

Class 1 Membership



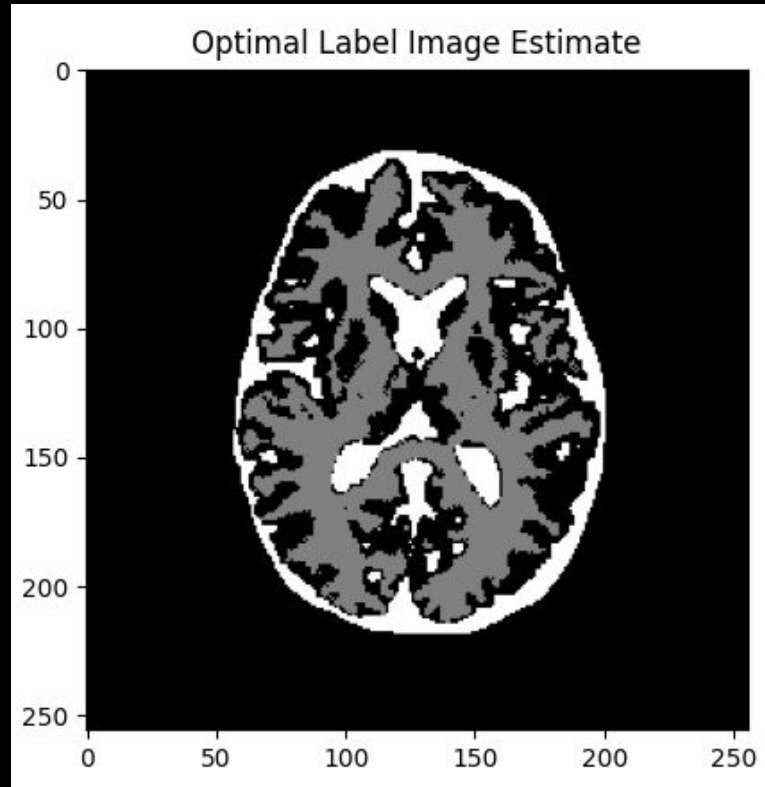
Class 2 Membership



Class 3 Membership



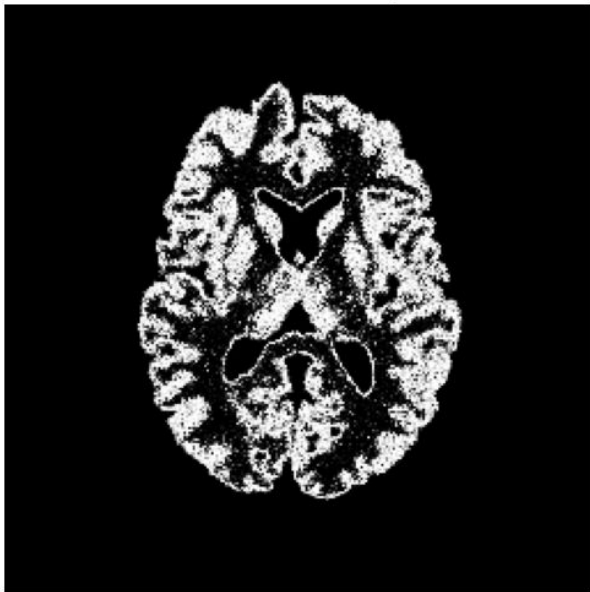
Q2e (iii)



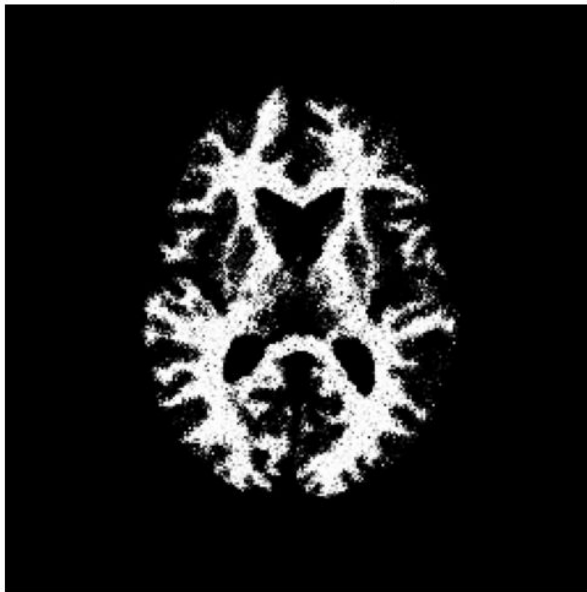
Q2e (iv)

Optimal Class Membership Estimate $\beta=0$

Class 1 Membership



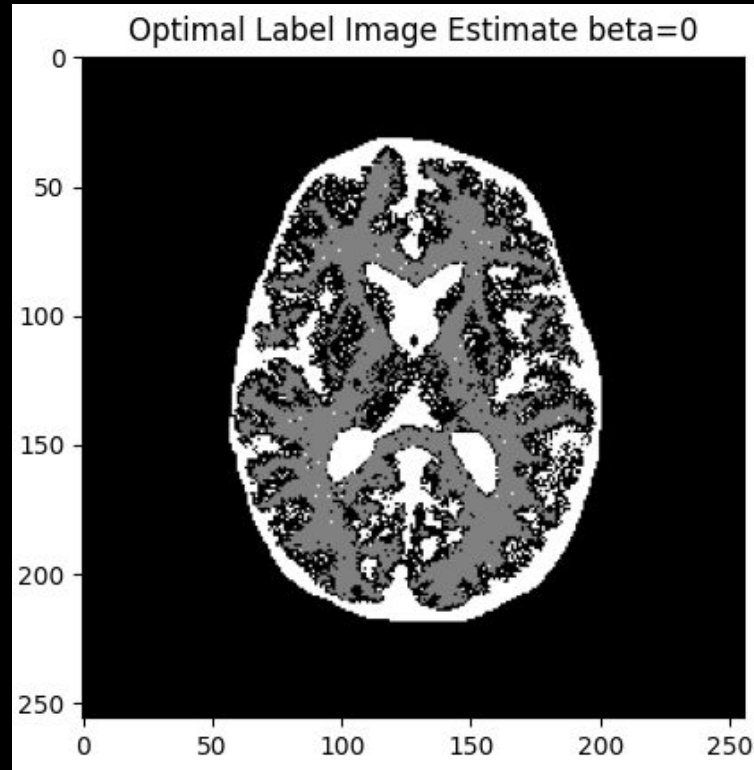
Class 2 Membership



Class 3 Membership



Q2 e (v)



Q2 f

Class 0 = 0.52287683

Class 1 = 0.63087137

Class 2 = 0.30621753

Thank You